Taut fillings

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Version dated 2026-03-20

Blue marks restored or substantially rewritten material, chiefly returning the proofs of Theorems 2 and 3 to the fuller version.
Red marks local corrections of notation or wording.

Abstract

Sleator, Tarjan, and Thurston asked: given a triangulation σ of the 2-sphere, what is the minimum number of tetrahedra needed to extend σ to a triangulation of the ball? Call this minimum $\mathrm{tetvol}(\sigma)$. Let X be the associated integral 2-cycle, and let $\mathrm{Zvol}(\sigma)$ be the minimum L_1-norm of an integral 3-chain M with $\partial M = X$. Certainly $\mathrm{tetvol}(\sigma) \geq \mathrm{Zvol}(\sigma)$. We show here that equality holds, and in fact any optimal filling M has support an anyroot stickerball with boundary σ: a clean simplicial triangulation of the 3-ball admitting a monotone shelling beginning with any prescribed tetrahedron. This complex is a flag complex: every clique in its 1-skeleton occurs as a simplex. The key to the proof is the general fact that any optimal filling of an integral n-cycle splits under disjoint union, connected sum, and more generally what we call almost disjoint union, where summands are supported on sets that overlap in at most $n+1$ vertices.

1 Overview

Let $\Delta = \Delta(\Omega)$ be the $|\Omega|-1$-simplex with vertices Ω, viewed as the abstract simplicial complex consisting of all subsets of Ω. Let $C_n = C_n(\Omega) = C_n(\Delta(\Omega), \mathbf{Z})$ and $Z_n = Z_n(\Omega) = Z_n(\Delta(\Omega), \mathbf{Z})$ be its integral n-chains and

*Zili Wang is supported by the National Natural Science Foundation of China (Grant No. 12301432)

n-cycles. A *filling* of an n-cycle $X \in Z_n$ is any $n{+}1$-chain $M \in C_{n+1}$ with $\partial M = X$. (Here and throughout we'll assume $n \geq 1$.) Let $\mathrm{Zvol}(X)$ be the minimum L_1-norm of a filling of X. Call $M \in C_{n+1}$ *taut* if it is an optimal filling of its boundary, i.e., if $|M| = \mathrm{Zvol}(\partial M)$,

For $X \in C_n$ let $\mathrm{Vert}(X)$ be the set of all the vertices of all the n-simplices to which X assigns non-0 weight. For any taut filling M of X we have $\mathrm{Vert}(M) = \mathrm{Vert}(X)$. (Cf. Proposition 4 below.) This is why we write C_n and Z_n without reference to Ω.

Call an n-cycle $X + Y \in Z_n$ an *almost disjoint union* if

$$|\mathrm{Vert}(X) \cap \mathrm{Vert}(Y)| \leq n + 1.$$

This notion generalizes both disjoint union and connected sum along an n-simplex. Ellison [2] showed the Zvol adds under almost disjoint union: $\mathrm{Zvol}(X + Y) = \mathrm{Zvol}(X) + \mathrm{Zvol}(Y)$. This means that *some* taut filling M of X splits into a sum $M = M_X + M_Y$ of taut fillings of X, Y. Here we amplify Ellison's Corollary 2 to show (Theorem 1) that when $n \geq 2$, *any* taut filling of $X + Y$ splits.

In place of integral chains we can use chains with coefficients in $\mathbf{Q}$ or $\mathbf{R}$. Evidently $\mathrm{Rvol} \leq \mathrm{Qvol}$. But computing Rvol is a linear programming problem with rational coefficients, so in fact $\mathrm{Qvol} = \mathrm{Rvol}$. Ellison used LP duality to prove that Qvol adds under almost disjoint union. Now for $n \geq 2$ we get the stronger result (Corollary 1) that taut $\mathbf{Q}$-fillings split: Multiplying a taut $\mathbf{Q}$-filling M of X by a common denominator q yields a taut $\mathbf{Z}$-filling qM of qX: Split the $\mathbf{Z}$-filling qM and divide by q to get a splitting of the $\mathbf{Q}$-filling M.

Now let σ be a simplicial triangulation of S^2; let $X(\sigma) \in Z_2(\Delta(\mathrm{Vert}(\sigma)), \mathbf{Z})$ be either one of the 2-cycles that arises from σ by orienting its 2-simplices; and write $\mathrm{Zvol}(\sigma) = \mathrm{Zvol}(X(\sigma))$. Let $\mathrm{tetvol}(\sigma)$ be the minimum number of 3-simplices required to extend σ to a simplicial triangulation τ of B^3. We will show that

$$\mathrm{Zvol}(\sigma) = \mathrm{tetvol}(\sigma).$$

Certainly $\mathrm{Zvol} \leq \mathrm{tetvol}$, because any extension τ induces a filling of $X(\sigma)$. Theorem 2 states that any taut filling M of $X(\sigma)$ arises in this way. Theorem 3 strengthens this to say that its support is an anyroot stickerball: it admits a monotone ball shelling beginning with any prescribed tetrahedron. The final theorem says that the same complex is flag: every clique in its 1-skeleton occurs as a simplex. The proof of Theorem 2 relies on Theorem 1.

2 Background

Our interest in Zvol stems from the work of Sleator, Tarjan, and Thurston [5]. They observed that coning from a vertex of maximal degree shows that a v-vertex triangulation σ of S^2 has tetvol $\leq 2v - 4 - \text{maxdeg}(\sigma)$. If $v \geq 13$ $\text{maxdeg}(\sigma) \geq 6$, so tetvol $\leq 2v - 10$. They produced examples for which tetvol $= 2v - 10$, provided v is larger than some unspecified bound, and conjectured that such examples should exist for any $v \geq 13$. Their approach was to realize σ as an ideal hyperbolic polyhedron with volume $2vV_0 - O(\log(v))$, where V_0 is the volume of an equilateral ideal tetrahedron, which is maximal. This implies tetvol $\geq 2v - O(\log(v))$. From there they bootstrapped their way up to showing that tetvol $= 2v - 10$. They suggested [5, p. 697] that in fact Qvol $= 2v - 10$, which would immediately imply tetvol $= 2v - 10$. Mathieu and Thurston [3] produced a class of examples, different from the examples of Sleator, Tarjan, and Thurston, for which they could show that Qvol $= 2v - 10$, still under the assumption that v is sufficiently large. In [1] we produced examples for any $v \geq 13$ with Qvol $= 2v - 10$, confirming the conjecture of Sleator, Tarjan, and Thurston. Theorem 2 here tells us that whenever we have Qvol $= 2v - 10$, or just Zvol $= 2v - 10$, any taut filling must arise from a triangulation of the ball.

About Qvol versus Zvol. In [1] we described triangulations of S^2 with Qvol $<$ Zvol. In all such examples that we know of with maxdeg ≤ 6, the gap between Qvol and Zvol is < 1, so that $\lceil \text{Qvol} \rceil =$ Zvol. Since Qvol and Zvol add under connected sum, the gap between them can be arbitrarily large (Ellison [2]), but taking the connected sum pushes maxdeg above 6.

3 Taut fillings

We can think of an n-chain $X \in C_n$ as a multiset of non-cancelling oriented simplices:

$$X = \sum_{t \in X} t.$$

In this sum t denotes an oriented simplex

$$t = [x_0, \ldots, x_n] = [x_{\pi(0)}, \ldots, x_{\pi(n)}], \quad \pi \text{ an even permutation},$$

and each unoriented simplex contributes a number of terms corresponding to its multiplicity.

The size of M as a multiset is its L_1-norm $|M|$. Write $U \subset M$ if U is a sub-multiset of M. This happens just when $|M| = |U| + |M - U|$.

Proposition 1. *If M is taut and $U \subset M$ then U is taut.*

Proof. This is clear, but let's drag it out. If U is not taut, there is a smaller filling V of ∂U. Take $M' = M - U + V$. In terms of multisets, this means that we substitute V for U, and then do any required cancellation of oppositely oriented simplices of $M - U$ and V. We have

$$\partial M' = \partial M - \partial U + \partial V = \partial M$$

and

$$|M'| \leq |M - U| + |V| = |M| - |U| + |V| < |M|,$$

contradiction. $\square$

4 Coning

For $x \in \Omega$, $U \in C_n$ let

$$\mathrm{nbhd}(x, U) = \sum_{t \in U : x \in \mathrm{Vert}(t)} t,$$

$$\deg(x, U) = |\mathrm{nbhd}(x, U)|,$$

$$\mathrm{maxdeg}(U) = \max_x \deg(x, U).$$

The *cone from x to U* is the $(n+1)$-chain

$$\mathrm{cone}(x, U) = \sum_{t \in U} \mathrm{adj}(x, t),$$

consisting of all the non-trivial oriented $(n+1)$-simplices $\mathrm{adj}(x, t) = [x x_0 \ldots x_n]$ obtained by adjoining x to $t = [x_0 \ldots x_n] \in U$. If $x \in \mathrm{Vert}(t)$ then t doesn't contribute to the sum, so $|\mathrm{cone}(x, U)| = |U| - \deg(x, U)$.

If U is closed then $\partial\, \mathrm{cone}(x, U) = U$, so

Proposition 2. *For any $X \in Z_n$*

$$\mathrm{Zvol}(X) \leq |X| - \mathrm{maxdeg}(X) \qquad \square$$

If $U \in Z_n$ and $x \notin \mathrm{Vert}(U)$ then $\deg(x, U) = 0$ and $|\mathrm{cone}(x, U)| = |U|$. In this case we call $\mathrm{cone}(x, U)$ a *complete cone*. By Proposition 2 a non-trivial complete cone is not taut. (This is a long-winded way of saying that instead of coning from $x \notin U$, we could have coned from some $x \in U$.) Thus:

Proposition 3. *If M is taut it contains no non-trivial complete cone.* $\square$

Call $x \in \mathrm{Vert}(M) \setminus \mathrm{Vert}(\partial M)$ an *internal vertex* of M. If x is internal to M then $\mathrm{nbhd}(x, M)$ is a complete cone, so:

Proposition 4. *If M is taut then it has no internal vertices.* $\square$

5 Almost disjoint unions

Recall that if $X, Y \in Z_n$ we call $X + Y$ an almost disjoint union if

$$|\mathrm{Vert}(X) \cap \mathrm{Vert}(Y)| \leq n + 1.$$

The most interesting special case is a connected sum, where $t = [c_0, c_1, \ldots, c_n]$ occurs once in X and $-t = [c_1, c_0, \ldots, c_n]$ occurs once in Y. For example, if $n = 1$ with X a cycle of length p and Y a cycle of length q the connected sum $X + Y$ is a cycle of length $p + q - 2$. Here Zvol adds, because $\mathrm{Zvol}(X) = p - 2$, $\mathrm{Zvol}(Y) = q - 2$, and

$$\mathrm{Zvol}(X + Y) = (p + q - 2) - 2 = \mathrm{Zvol}(X) + \mathrm{Zvol}(Y).$$

But not every filling of $X + Y$ splits, because a taut filling may contain any 2-simplex with vertices in $\mathrm{Vert}(X + Y)$.

We want to show that fillings do split when $n \geq 2$.

For $A \subset \Omega$, $p \in A$ define

$$\pi_{A,p} : \Omega \to A$$

$$\pi_{A,p}(x) = x \textbf{ if } x \in A \textbf{ else } p$$

and let

$$K_*(A, p) : C_*(\Omega) \to C_*(A)$$

be the induced chain map. This is a projection of $C_*(\Omega)$ onto $C_*(A)$.

Let A, B be finite vertex sets sharing the vertices $C = A \cap B$, and let $c = |C|$. Observe that $C_c(C)$ and $Z_{c-1}(C)$ are trivial, because $\Delta(C)$ contains no c-simplex, and (up to orientation) only one $c-1$-simplex, with nothing to cancel its boundary.

Let
$$C_*(A,B) = C_*(A) \oplus C_*(B).$$
For $p, q \in C$ define the mapping
$$g_*(p,q) = K_*(A,p) \oplus K_*(B,q) : C_*(A \cup B) \to C_*(A,B).$$

Proposition 5. *Let $(X,Y) \in C_n(A,B)$. If $|A \cap B| \leq n$ we can recover (X,Y) from $X+Y$. If $(X,Y) \in Z_n(A,B)$ this holds also when $|A \cap B| = n+1$.*

Proof. Let $C = A \cap B$. We can assume $|C| \geq 1$. (Add a brand new point to C if necessary.) We claim that for any $p, q \in C$ (not necessarily distinct) we have
$$g_n(p,q)(X+Y) = (X,Y).$$
Indeed,
$$K_n(A,p)(X+Y) = X + K_n(A,p)(Y).$$
The second term belongs to $C_n(C)$, which is trivial when $|C| \leq n$. If $Y \in Z_n(B)$ the second term belongs to $Z_n(C)$, which is trivial when $|C| \leq n+1$.
$\square$

Theorem 1. *If $|A \cap B| \leq n+1$, for all $(X,Y) \in Z_n(A,B)$ we have*
$$\mathrm{Zvol}(X+Y) = \mathrm{Zvol}(X) + \mathrm{Zvol}(Y).$$
And as long as $n \geq 2$, for any $M \in \mathrm{taut}(X+Y)$ we have $M = M_X + M_Y$ with $M_X \in \mathrm{taut}(X)$, $M_Y \in \mathrm{taut}(Y)$.

Note. Ellison [2] proved the additivity of Zvol. What's new here is the splitting of taut fillings when $n \geq 2$. This result resembles Theorem 6.2 of Pournin and Wang [4] about flip paths. Like theirs, our proof uses a variation on the normalization technique of Sleator, Tarjan, and Thurston [5, Lemma 7]. It would be nice to fit these results under one roof.

Proof. Again let $C = A \cap B$. We can assume $|C| = n+1$, as this is the hardest case. And we might as well go ahead and take $n = 2$, $|C| = 3$, as this case illustrates all the issues.

Take any $M \in \mathrm{taut}(X+Y) \subset C_3(A \cup B)$. Pick distinct points $p, q \in C$, and let
$$(M_X, M_Y) = g_3(p,q)(M).$$

(For now we'll suppress the dependence of M_X, M_Y on p, q.)

Because $g_*(p,q)$ is a chain map we have $\partial_3 M_X = X$, $\partial_3 M_Y = Y$:

$$(\partial_3 M_X, \partial_3 M_Y) = \partial_3(g_3(p,q)(M)) = g_2(p,q)(\partial_3 M) = g_2(p,q)(X+Y) = (X,Y).$$

We want to show that under the map $g_3(p,q)$ any $t \in M$ dies either in M_X or M_Y, because then we'll get additivity of Zvol from

$$\mathrm{Zvol}(X+Y) = |M| \geq |M_X| + |M_Y| \geq \mathrm{Zvol}(X) + \mathrm{Zvol}(Y) \geq \mathrm{Zvol}(X+Y).$$

Let's call a 3-simplex a 'tet' for short. Say that a tet $t \in M$ has type $CCXY$ if $t = \pm[c_1 c_2 x_1 y_1]$ for $c_1, c_2 \in C$, $x_1 \in A \setminus C$, $y_1 \in B \setminus C$. Similarly for types $XXXX$, $CXXX$, $CXXY$, $XXXY$, etc. The first two are pure X cases, meaning that they live in $C_3(A)$; the last two are hybrid cases.

Any $XX..$ tet dies in M_Y; any $YY..$ tet dies in M_X.

The remaining cases to check are are the pure cases $CCCX$, $CCCY$, and the hybrid case $CCXY$. A $CCCX$ tet $t = \pm[c_1 c_2 c_3 x_1]$ dies in M_Y because $K_3(B,q)(t) = [c_1 c_2 c_3 q]$, which vanishes because $q \in C = \{c_1, c_2, c_3\}$ produces a repeated vertex. Likewise a $CCCY$ tet dies in M_X. The more interesting case is $CCXY$: The key is that since $|C| = 3$, $\{c_1, c_2\}$ cannot be disjoint from $\{p, q\}$, so it dies on one side or the other.

So

$$|M| = |M_X(p,q)| + |M_Y(p,q)|,$$

and Zvol adds. Note how we're now emphasizing the possible dependence of M_X, M_Y on p, q. When $n = 1$ the choice of p, q can indeed make a difference: We can get a different pair (M_X, M_Y) if we switch p and q. (Think about the connected sum of two cycle graphs.)

But when $n \geq 2$ we will now show that all tets are pure X or pure Y, so M_X consists of all the pure X tets, and ditto for M_Y. This will make $M = M_X + M_Y$ with $M_X \in \mathrm{taut}(X)$, $M_Y \in \mathrm{taut}(Y)$.

Again our test case is $n = 2$.

The key observation is that, now that we know that every tet dies on one side or the other, we know that no tet can die on both sides, because that would make $|M| > \mathrm{Zvol}(X) + \mathrm{Zvol}(Y)$.

An $XXYY$ tet would die on both sides, so there can be none of these.

Any $pqXY$, $pXXY$, or $qXYY$ would die on both sides, and p, q are arbitrary, so this rules out all $CCXY, CXXY, CXYY$.

The only remaining hybrids are $XXXY$ and $XYYY$. Let's rule out $XXXY$, leaving $XYYY$ to symmetry.

Fix any $y_0 \in B \setminus C$, and suppose there is some $XXXy_0$ tet in M. Let $U \subset M$ consist of all such $XXXy_0$ tets. We claim that $y_0 \notin \mathrm{Vert}(\partial U)$. For let s be any 2-simplex of the form XXy_0, the only kind of 2-simplex containing y_0 that could belong to ∂U. Any $t \in M$ of which s or $-s$ is a face must be a hybrid tet, and the only possibility is $XXXy_0$, so $t \in U$. Because $\partial M = 0$ the signed multiplicity of s vanishes in ∂M, hence also in ∂U, because the tets of M with $\pm s$ as a face all belong to U. So $y_0 \notin \mathrm{Vert}(\partial U)$, making U a complete cone on y_0, contradicting 3. So there is no such $XXXy_0$ tet, ruling out $XXXY$, and with it $XYYY$.

So there are no hybrid tets, meaning that M splits, as claimed. $\square$

The splitting of taut fillings depends on the assumption $n \geq 2$. Let's consider what goes wrong when $n = 1$. Here the hybrid types are CXY, XXY, XYY. Neither pXY nor qXY dies on both sides, so we can't rule out CXY. Failing that, the XXy_0 tets needn't form a complete cone, because $[x_0x_1y_0)$ can continue across $[x_1y_0]$ to $-[px_1y_0]$, so we can't rule out XXY either.

The foregoing proof works equally well over $\mathbf{Q}$, providing we permit ourselves to work with fractional multisets. Alternatively, we can clear denominators as in the introduction above. Either way, we have:

Corollary 1. Qvol *adds under almost disjoint union, and for $n \geq 2$ taut $\mathbf{Q}$-fillings split.* $\square$

6 Filling a triangulation of the 2-sphere

In this section we again view an integral 3-chain M as a signed multiset of oriented tetrahedra. We call M *simplicial* if, after forgetting orientations, no tetrahedron occurs with multiplicity larger than 1. In that case its support determines a pure simplicial 3-complex. We call M *clean* if this associated complex is clean: triangular faces are incident to at most two tetrahedra, and the links of vertices and edges are connected. This is the usual normal pseudomanifold condition, with boundary allowed, specialized to dimension 3.

Theorem 2. *Let σ be a simplicial triangulation of S^2, and let M be a taut filling of σ. Then M is clean, and arises from a simplicial triangulation of B^3.*

Proof.

Let v, e, f count the vertices, edges, and faces of σ. We have $e = 3v - 6$, $f = 2v - 4$. (Check: $3f = 2e$, $v - e + f = 2$.)

Consider a counterexample pair (σ, M) for which $\mathrm{Zvol}(\sigma) = |M|$ is minimal, among all taut integral fillings of simplicial 2-spheres. Thus the induction hypothesis is the full statement of this theorem for every smaller taut filling: the smaller filling is simplicial, clean, and gives a triangulation of B^3. Obviously $v > 4$. Because taut fillings split under connected sum, we can assume that σ is prime (not a connected sum along a triangle), and in particular (this is all we will need) that there is no vertex of degree 3.

Call a tet $t \in M$ *eligible* if it shares two faces with σ. It cannot share more, since σ has no vertex of degree 3. Since

$$|M| = \mathrm{Zvol}(\sigma) \leq f - \mathrm{maxdeg}(\sigma)$$

there must be at least $\mathrm{maxdeg}(\sigma)$ disjointly eligible tets, but we will only need two: one with faces $s_1, s_2 \in \sigma$, the other with disjoint faces $s_3, s_4 \in \sigma$. The s_3, s_4 tet cannot also have s_1 or s_2 as a face, as then there would be a degree-3 vertex in σ. So for any face s of σ there is an eligible tet without s as a face.

When we remove an eligible tet t, we get a taut filling $M - t$ of a 2-complex σ_t, where σ_t is either (1) a triangulation of S^2, or (2) the almost disjoint union of two triangulations σ_1, σ_2 of S^2 that are joined along an edge of t. We claim first that $M - t$ is simplicial for every eligible tet t. In case (1), $M - t$ is itself a taut filling of the smaller sphere σ_t, so by minimality it is clean, hence simplicial. In case (2), $M - t$ is a taut filling of an almost disjoint union. By Theorem 1 it splits as $M - t = M_1 + M_2$, where each M_i is a taut filling of σ_i. Each M_i has smaller volume, so by minimality each is clean and simplicial. Hence $M - t$ is simplicial. In this case $M - t$ need not be clean as a single complex, because the link of the common edge shared by σ_1 and σ_2 is disconnected; only the two summands are clean.

It follows that M itself is simplicial. Suppose some tetrahedron u occurs in M with multiplicity greater than 1 (after forgetting orientation). Choose an eligible tetrahedron $t \neq u$; this is possible because we have at least two eligible tetrahedra. Then $M - t$ still contains u with multiplicity greater than 1, contradicting the simpliciality of $M - t$. Thus every tetrahedron of M has multiplicity 1.

The crux is now to show that this simplicial chain is clean.

Now:

1. M is a pseudomanifold. For suppose some 2-simplex s occurs more than twice as the boundary of a 3-simplex of M. If $s \notin \sigma$, it must occur at least twice plus and twice minus; after removing any eligible tet it still occurs either twice plus or twice minus in $M - t$ in case (1), or in one or the other of M_1, M_2 in case (2), contradicting minimality. If $s \in \sigma$, it occurs at least twice plus in M: remove some eligible tet that does not have s as a face to get a contradiction.

2. M has no edge $\{a, b\}$ whose link is disconnected. For the link is a 1-dimensional complex whose edges correspond to tets of M that contain $\{a, b\}$. The only way a component of the link can have fewer than three vertices is if it consists of a single edge $\{c, d\}$, corresponding to a tet $t = \{a, b, c, d\} \in M$. In this case $\{a, b\}$ must be an edge of σ, and $\{a, b, c\}, \{a, b, d\}$ the faces of σ that are adjacent along $\{a, b\}$. These are the only faces of σ that contain $\{a, b\}$, so no other component of the link can reduce to a single edge. If the link is disconnected, removing any eligible tet other than t will leave it disconnected, contradicting minimality.

3. M has no vertex whose link is disconnected. For if $\operatorname{link}(\{a\}, M)$ is not connected, the only way removing a single tet t can render the link connected is if one component of the link has a single 2-simplex $\{b, c, d\}$ and $t = \{a, b, c, d\}$. In this case $\{a, b, c\}$, $\{a, b, d\}$, $\{a, c, d\}$ must all be faces of σ, making a a vertex of σ of degree 3, contradiction.

Hence M is clean. Therefore its support is the desired simplicial triangulation of B^3. $\square$

Thanks to this theorem, when M is a taut filling of σ we may write $\tau = \operatorname{supp} M$ and regard τ as a simplicial triangulation of B^3 with boundary σ.

7 Stickerballs

A *monotone shelling* of a clean finite 3-complex τ with nonempty boundary is an ordering $(t_1, \dots, t_{|\tau|})$ of its tets such that, for each k, the subcomplex τ_k generated by $\{t_1, \dots, t_k\}$ is again clean with nonempty boundary, and t_k is attached to τ_{k-1} along one or two boundary triangles. Equivalently, the usual three-face attachment, which would decrease the number of boundary vertices, is disallowed.

A *stickerball* is a clean finite 3-complex with nonempty boundary, equipped with such a monotone shelling. By the standard PL interpretation of shellings, a stickerball is a triangulated 3-ball. A stickerball is *rooted at* a tetrahedron t if it has a monotone shelling beginning with t. It is an *anyroot stickerball* if every tetrahedron can serve as the root. Thus "anyroot stickerball" is stronger and more specific than "anyroot shellable": the object is a ball rather than a sphere, and the shellings are monotone ball shellings, with attachments along one or two boundary triangles.

Theorem 3. *If σ is a simplicial triangulation of S^2, and M is any taut integral filling of σ, then $\tau = \operatorname{supp} M$ is an anyroot stickerball with boundary σ.*

Proof. By Theorem 2, $\tau = \operatorname{supp} M$ is already a clean simplicial triangulation of B^3 with boundary σ. Fix a tetrahedron $r \in \tau$. We run the deletion argument of Theorem 2 with r held back until the end. Whenever an eligible tet is to be removed, choose one different from r; the counting above supplies at least two eligible tets, so this is possible unless r has already been isolated as the last tet of the current component. If removal disconnects the boundary into two spheres joined along an edge, Theorem 1 splits the filling into two taut fillings. First shuck the component not containing r, and then continue recursively with the component containing r. This produces a deletion order whose final tetrahedron is r. Reversing the order gives a monotone shelling beginning with r. Since r was arbitrary, τ is an anyroot stickerball. $\square$

8 Flag complex

A simplicial complex is a *flag complex* if every clique in its 1-skeleton occurs as a simplex. (Through gritted teeth we say that the complex 'is flag'.) A simplicial triangulation σ of S^2 is flag just if it is *prime* (not the connected sum of smaller complexes) and not the boundary of a tetrahedron.

We can decompose any σ as the connected sum of prime components σ_i. (Of course, to reconstruct σ we need more than this, we need to know just how they are connected.) From the point of taut fillings these components are independent: If M is a taut filling of σ, then $\tau = \operatorname{supp} M$ is a union of the supports τ_i of taut fillings of the prime components σ_i.

Now we want to show that, whether or not σ itself is prime, the support of any taut filling of σ is a flag complex.

Theorem 4. *If σ is a simplicial triangulation of S^2, and M is any taut filling of σ, then $\tau = \operatorname{supp} M$ is a flag complex.*

Proof. The goal here is to show that none of these three taboo configurations occurs in τ:

1. The three edges of triangle $\{A, B, C\}$ without the 2-simplex ABC.
2. The six edges of the K_4 $\{A, B, C, D\}$ without the 3-simplex $ABCD$.
3. The ten edges of the K_5 $\{A, B, C, D, E\}$.

If we can rule out 1 and 2 then 3 will follow, because along with the K_5 we would have the five associated 3-simplices: Together these would form an S^3, which is not possible as a subcomplex of the topological 3-ball τ.

To rule out 1 and 2, the strategy will be to consider a minimal counterexample, and find a modification that would produce a smaller counterexample.

If σ has a vertex of degree 3, it is a connected sum with one summand a single tetrahedron. We can pull off the vertex from σ, while removing the associated tet from τ, to produce a smaller counterexample.

If σ has no vertex of degree 3, there are at least maxdeg ≥ 4 disjointly eligible tet. Removing any one of these tets performs an 'edge flip' on the boundary σ, which either (a) remains a triangulation of the 2-sphere, or (b) becomes an almost disjoint union where two such triangulations are joined along an edge. We will show that some choice for the flip does not remove the taboo configuration. In case (a) we have a smaller counterexample already. In case (b) the key is that if a K_3 or K_4 occurs in the almost disjoint union, it occurs on one side or the other, also giving a counterexample.

Configuration 1: When we remove an eligible tet from τ to flip an edge of σ, the only edge that disappears from τ is the edge that gets flipped. There are at least four choices of the edge to flip, and three edges we must avoid flipping to preserve the taboo K_3.

Configuration 2: Now there are six edges we must avoid flipping. Having dealt with configuration 1, we know that along with the six edges of $\{A, B, C, D\}$, τ must contain all four of the unoriented faces ABC, ABD, ACD, BCD. At most two of these can be boundary faces, because σ has no vertex of degree 3. So at most five of the K_5 edges are on the boundary. By treating the octahedron separately as a special case, we can assume maxdeg ≥ 5, so we have at least five boundary edges we can flip. The only potential problem is if five of the six K_4 edges are on the boundary. Say the interior edge is AD, so that the K_4 includes the faces ABC, BCD. If an eligible tet contains neither of these as one of its two boundary faces, we are fine. But at most two of our disjointly eligible tets can contain one of the faces ABC, BCD, leaving us at least maxdeg $- 2$ other flipping options. $\square$

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